ACICLOVIR

Hydrate — crystal nephropathy

Dose on ideal weight

CHEMICAL NAME 2-amino-9-[(2-hydroxyethoxy)methyl]-1,9-dihydro-6H-purin-6-one (acyclovir)

CLASS Nucleoside analogue antiviral (guanosine analogue)

TRADE NAMES Zovirax · Acyclovir · Aciclovir cream

THERAPEUTIC USES / INDICATIONS

- Herpes simplex encephalitis (IV, urgent).
- Genital and oral herpes (HSV); herpes keratitis (eye ointment).
- Varicella (chickenpox) in adults/immunocompromised; herpes zoster (shingles).
- Prophylaxis of HSV/VZV in immunocompromised patients; neonatal herpes.

ADMINISTRATION

- **PO** with or without food; 5 times daily for herpes zoster.
- **IV:** infuse over at least 1 h (rapid infusion → crystal nephropathy); ensure good hydration.
- **Topical** cream 5 times daily; **eye ointment** 5 times daily.
- IV solution is alkaline — avoid extravasation.

DOSAGES

- **Genital herpes (first episode):** 400 mg PO three times daily for 5–10 days.
- **Shingles:** 800 mg PO five times daily for 7 days (start within 72 h).
- **HSV encephalitis:** 10 mg/kg IV every 8 h for 14–21 days.
- **Children IV (encephalitis):** 3 months–12 yr 500 mg/m² every 8 h; neonates 20 mg/kg every 8 h.
- Reduce in renal impairment; obese patients use ideal body weight.

PHARMACOKINETICS

- **Peak:** PO 1.5–2 h. Oral bioavailability only 15–30%.
- Good CSF penetration (~50%).
- Excreted unchanged in urine. **t½:** 2–3 h (up to 20 h in renal failure).

PHARMACODYNAMICS

- Phosphorylated by viral thymidine kinase, then by host enzymes to aciclovir triphosphate → inhibits **viral DNA polymerase** and terminates DNA chain.
- Selective for infected cells → low toxicity.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, diarrhoea, headache.
- Phlebitis (IV).
- Topical: stinging.

ADVERSE EFFECTS (SERIOUS)

- Acute kidney injury (crystal nephropathy) with IV, dehydration.
- Neurotoxicity: confusion, tremor, hallucinations, seizures (renal impairment, elderly).
- Bone-marrow suppression (rare).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to aciclovir or valaciclovir.
- **Caution:** renal impairment, dehydration, elderly, nephrotoxic drugs.

KEY INTERACTIONS

- **Nephrotoxic drugs** (aminoglycosides, ciclosporin, NSAIDs): ↑ AKI risk.
- **Probenecid, cimetidine:** ↑ levels.
- **Mycophenolate:** ↑ levels of both.

NURSING MANAGEMENT

- Suspected encephalitis: do not delay the first IV dose.
- Make sure adequate hydration (IV fluids/oral); monitor urine output.
- Monitor creatinine every 2–3 days on IV therapy.
- Watch for confusion or tremor.
- Isolate chickenpox/disseminated zoster (airborne + contact).

PATIENT EDUCATION

- Start tablets as soon as symptoms begin.
- Drink plenty of fluids.
- Avoid sexual contact during outbreaks; condoms reduce spread.
- Cover shingles rash; avoid pregnant women and newborns until crusted.

KEY POINTS

- Guanosine analogue; activated by viral thymidine kinase.
- Shingles 800 mg 5×/day.
- HSV encephalitis 10 mg/kg IV every 8 h.
- Hydrate — crystal nephropathy; neurotoxicity in renal failure.